SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – PROMPT ENGINEERING CHALLENGE

Course Code:	CSA65	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO2 – Precision (BL2, BL3)	**Assessment:**	Assessment Tool 1 – Prompt Engineering
Weightage:	20% of CIA	**Total Marks:**	2 * 50 = 100 (2 Questions)
CO2 Statement:	*Apply prompt engineering techniques and utilize pre-trained Large Language Models through modern AI frameworks and APIs to solve real-world problems (BL2, BL3)*		
Student Name:	GUDIPATI PRANATHI	**Reg. No.:**	192472237
Section / Batch:	CSE-AI/2024	**Date:**	26-10-2026

Instructions to Students
• Answer 2 questionss.Each questions carries 50 marks..Total100 Marks.
1. Design appropriate prompts for the given tasks using suitable prompt engineering techniques.
2. Select and apply the appropriate prompting strategy (Zero-shot, One-shot, Few-shot, Chain-of-Thought, or Structured Prompting) based on the given scenario.
3. Include clear instructions, relevant context, necessary constraints, and the expected output format while designing each prompt.
4. Evaluate and refine the designed prompts to improve their clarity, effectiveness, and quality by applying prompt engineering best practices.
5. Design prompts for a variety of real-world applications, such as text generation, summarization, translation, question answering, code generation, and conversational AI.
6. Justify the designed prompt by briefly explaining the selected prompting technique and how it helps achieve the desired output.

Code of Conduct
I G.PRANATHI [192472237] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G.PRANATHI

SECTION A – PROMPT ENGINEERING CHALLENGE

1. Design a Zero-shot prompt to summarize a news article in 100 words.
2. Create a One-shot prompt to translate an English paragraph into Tamil using one example.
3. Design a Few-shot prompt to classify movie reviews as Positive, Negative, or Neutral.
4. Write a Chain-of-Thought prompt to solve a multi-step mathematical problem.
5. Create a Structured Prompt to generate a professional email requesting leave from work.
6. Design a prompt to generate a Python program that checks whether a number is prime.
7. Write a prompt to generate SQL queries for retrieving student records from a database.
8. Create a prompt to summarize a research paper into key points.

9. Design a prompt to generate interview questions for a Software Engineer position.
10. Write a prompt to explain Artificial Intelligence to a school student using simple language.
11. Design a prompt to create a lesson plan for teaching Machine Learning to undergraduate students.
12. Create a prompt to generate a travel itinerary for a three-day trip to Chennai.
13. Write a prompt for a recipe recommendation system based on available ingredients.
14. Design a prompt for a customer support chatbot to answer product-related queries.
15. Create a prompt to generate social media captions for a new smartphone launch.
16. Write a prompt to generate a product description for an e-commerce website.
17. Design a prompt for performing sentiment analysis on customer feedback.
18. Create a prompt to perform grammar correction on a given paragraph.
19. Write a prompt to convert technical content into simple language for beginners.
20. Design a prompt to generate multiple-choice questions from a textbook chapter.
21. Create a prompt to generate a project abstract from a given project title.
22. Write a prompt to generate personalized study notes for engineering students.
23. Design a prompt to generate a professional resume based on candidate details.
24. Create a prompt to write a cover letter for a Software Developer position.
25. Write a prompt to generate Python code documentation from source code.
26. Design a prompt that explains a programming error and suggests possible fixes.
27. Create a prompt to generate unit test cases for a Python function.
28. Write a prompt to compare BERT, GPT, LLaMA, and Gemini in tabular format.
29. Design a prompt to explain the Transformer architecture with a suitable diagram description.
30. Create a prompt to compare Pre-training and Fine-tuning with suitable examples.
31. Write a prompt to explain the concept of Transfer Learning for beginners.
32. Design a prompt to explain the advantages of LoRA and PEFT in fine-tuning Large Language Models.
33. Create a prompt that demonstrates Function Calling for retrieving weather information.
34. Write a prompt using the OpenAI API to generate a customer support response.
35. Design a prompt using the Gemini API to summarize meeting minutes.
36. Create a prompt using the Hugging Face API to perform sentiment analysis.
37. Write a prompt to evaluate and improve a poorly written text generation prompt.
38. Design a prompt following Prompt Engineering Best Practices to generate educational content.
39. Create a prompt for an AI Career Guidance Assistant that recommends suitable career paths based on student interests.
40. Design a comprehensive prompt for an AI-powered Engineering Assistant that answers technical questions, explains concepts, and provides code examples with appropriate formatting.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technique Selection	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Output Effectiveness	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Prompt Craftsmanship	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Prompt-Output Documentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Design Rationale	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technique Selection (30%)	Output Effectiveness (25%)	Prompt Craftsmanship (20%)	Prompt-Output Documentation (15%)	Design Rationale (10%)	Total Marks (100)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

PROMPT ENGINEERING CHALLENGE – SOLUTION

14. Design a Prompt for a Customer Support Chatbot to Answer Product-Related Queries

1. AIM

To design a well-structured prompt for a Generative AI-based customer support chatbot that answers product-related queries accurately, provides personalized assistance, explains product features, and improves customer satisfaction using Large Language Models (LLMs).

2. PROMPT DESIGN

Prompt for Customer Support Chatbot

Role:

You are an intelligent, friendly, and professional AI customer support chatbot for an online shopping platform.

Objective:

Your main goal is to help customers understand products, compare items, check availability, learn about pricing, and resolve common product-related problems.

Instructions:

1. Product Information:
   Answer questions about product names, features, specifications, sizes, colors, materials, compatibility, and usage. Use only verified product information provided by the store.
2. Pricing and Availability:
   Provide the current product price, discounts, stock status, and delivery information only when reliable data is available. Never invent prices or availability.
3. Product Recommendations:
   Recommend products based on customer needs, preferences, budget, and intended usage. Explain why each recommendation is suitable.
4. Product Comparison:
   Compare two or more products based on price, features, quality, performance, and customer requirements. Present the differences clearly.

5. Returns and Refunds:
   Explain the store's return, replacement, refund, and warranty policies using the official policy information. Do not promise refunds without confirmation.
6. Understanding Customer Queries:
   Ask relevant follow-up questions when a customer's request is incomplete or unclear. Provide personalized responses whenever sufficient information is available.
7. Communication Style:
   Use simple English, short paragraphs, and bullet points. Remain polite, patient, respectful, and professional.
8. Accuracy and Safety:
   Do not generate false information, guess product specifications, or make unsupported claims. If information is unavailable, honestly inform the customer.
9. Privacy:
   Never ask customers to share passwords, OTPs, PINs, or complete payment card details.
10. Escalation:
    If the issue is complex, involves a payment dispute, or cannot be resolved by the chatbot, guide the customer to a human support representative.

Expected Response Format:

- Direct answer to the customer's question
- Relevant product details
- Additional guidance or recommendations, if needed
- Next step for the customer

Example Interaction:

Customer: I need a laptop for college under ₹50,000.

Chatbot: I can help you find a suitable laptop within your budget. For college use, I recommend looking for at least 8 GB RAM, an SSD, good battery life, and a processor suitable for your applications. Could you tell me whether you mainly need it for programming, online classes, gaming, or general use?

Final Goal:
Provide accurate, useful, and personalized customer support while building customer trust and improving the online shopping experience.

3. TECHNIQUES USED

Technique	Explanation	Example
Role-based prompting	Assigns a specific role to the AI.	AI acts as a customer support agent.
Instruction-based prompting	Provides clear rules and tasks.	Answer product questions using verified information.
Contextual prompting	Uses product details and customer requirements.	Recommends laptops based on budget and usage.
Few-shot prompting	Provides sample interactions to guide responses.	Demonstrates how to answer a laptop query.
Constraint-based prompting	Defines limits to prevent incorrect responses.	Never invent prices or warranty terms.

4. WORKING PROCESS

Step 1: Customer Query

Customer asks about a product, price, feature, or problem.

Step 2: Prompt Instructions

The chatbot follows its assigned role, rules, and response format.

Step 3: AI and LLM Processing

The language model interprets the question and generates a relevant response using available context.

Step 4: Response Validation

The response is checked against available product information and store policies.

Step 5: Customer Support Response

The chatbot provides an answer, asks a follow-up question, or escalates the issue.

5. SAMPLE INPUT AND OUTPUT

Example 1: Product Features

Input:

Customer: Does this smartphone support 5G?

Expected Output:

Chatbot: I can help you check that. Please share the smartphone model or product link so I can verify its network specifications.

Example 2: Product Recommendation

Input:

Customer: Suggest wireless earphones under ₹2,000.

Expected Output:

Chatbot: I can help you choose wireless earphones within ₹2,000. Consider battery life, sound quality, microphone performance, comfort, and warranty. If you share your priorities, I can recommend suitable options from the available products.

Example 3: Product Comparison

Input:

Customer: Which is better, a 128 GB phone or a 256 GB phone?

Expected Output:

Chatbot: A 128 GB phone may be sufficient for basic use, while a 256 GB phone is better if you store many photos, videos, games, or large applications. Your choice depends on your storage needs and budget.

Example 4: Return Policy

Input:

Customer: Can I return a damaged product?

Expected Output:

Chatbot: I'm sorry your product arrived damaged. Please check the store's damaged-item return policy and report the issue within the specified period. Keep your order details and product photos ready. I can guide you through the next steps using the store's official policy.

6. TECHNOLOGY CONCEPTS USED

Concept	Role in the chatbot	Real-world example
Artificial Intelligence (AI)	Enables the system to perform customer support tasks.	Automated shopping assistance.
Machine Learning (ML)	Helps models learn language patterns from training data.	Understanding different ways customers ask questions.
Deep Learning (DL)	Uses neural networks to process complex language patterns.	Understanding product questions written in natural language.
Generative AI	Generates human-like responses to customer queries.	Creating a helpful response to a complaint.
Large Language Model (LLM)	Interprets prompts and generates contextual answers.	Answering product specification questions.

These concepts are related hierarchically: AI is the broad field; ML is a subfield of AI; DL is a subfield of ML; and Generative AI uses various approaches, including deep learning. LLMs are deep-learning models commonly used for generative language tasks.

7. ADVANTAGES

1. Provides quick responses to customer queries.
2. Offers 24/7 automated assistance.
3. Reduces the workload of human support agents.
4. Improves product discovery and comparison.
5. Provides consistent responses based on store policies.
6. Helps improve customer satisfaction.
7. Supports personalized product recommendations.
8. Reduces misinformation through clear instructions and verified product data.

8. DESIGN RATIONALE

The prompt is designed with a clear role, objective, instructions, examples, response format, and safety constraints. Each component improves the chatbot's performance.

- Role definition ensures the AI behaves like a professional support agent.
- Detailed instructions cover common product-related queries.
- Examples demonstrate the expected communication style.
- Verified context helps prevent fabricated prices, specifications, and policies.
- Safety constraints protect customer privacy and reduce unsupported claims.
- Escalation rules ensure that complex problems reach human support staff.

9. RESULT

A detailed, structured prompt was designed for a Generative AI-based customer support chatbot. The prompt combines role-based, instruction-based, contextual, few-shot, and constraint-based prompting techniques to deliver accurate, clear, personalized, and reliable product-related assistance. It demonstrates how AI, ML, DL, Generative AI, and LLMs can support real-world customer service applications.

19. Write a Prompt to Convert Technical Content into Simple Language for Beginners

1. AIM

To design an effective prompt using Generative AI and Large Language Models (LLMs) to transform complex technical information into simple, beginner-friendly language while maintaining accuracy, meaning, and important technical details.

2. PROMPT DESIGN

Prompt for Simplifying Technical Content

Role:

You are an expert technical educator and content simplification specialist who helps beginners understand difficult technical concepts.

Objective:

Convert the given technical content into simple, clear, and understandable language for students and beginners who have little or no prior knowledge of the topic.

Instructions:

1. Understand the original technical content before rewriting it.
2. Use simple English, short sentences, and familiar words.
3. Explain technical terms using easy definitions.
4. Break complicated concepts into smaller, logical steps.
5. Use real-world examples and everyday comparisons to explain abstract ideas.
6. Preserve the original meaning, important facts, formulas, and technical accuracy.
7. Avoid unnecessary jargon, complex grammar, and lengthy paragraphs.
8. Use headings, bullet points, numbered steps, and tables where appropriate.
9. Explain how the concept works and why it is useful.
10. Include practical applications from daily life, education, or industry.
11. Identify common beginner misunderstandings and clarify them.
12. If the content contains formulas or code, explain each important part in simple language without changing its function.
13. Do not invent facts or introduce unsupported claims.
14. If the original content is unclear or incomplete, identify the uncertainty instead of guessing.
15. Adjust the explanation to a beginner's level while retaining essential technical terminology.

Required Output Format:

A. Topic Title

B. Simple Introduction

C. Explanation in Beginner-Friendly Language

D. Important Technical Terms and Their Meanings

E. Step-by-Step Working Process

F. Real-World Example or Analogy

G. Practical Applications

H. Advantages and Limitations, where relevant

I. Common Beginner Mistakes

J. Short Summary

K. Three Simple Review Questions with Answers

Technical Content to Simplify:

"Machine learning is a subset of artificial intelligence that enables computer systems to identify patterns in data and make predictions or decisions without being explicitly programmed for every individual task. It uses algorithms to learn from training data and improve performance on a task."

Final Requirement:

Produce an accurate, well-organized, beginner-friendly explanation that helps a student understand the concept, remember its key ideas, and apply the knowledge in real-life situations.

3. TECHNIQUES USED

Technique	Explanation	Application
Role-based prompting	Assigns a specific role to the AI.	AI acts as a technical educator.
Instruction-based prompting	Gives detailed instructions for the task.	Uses simple language and short sentences.
Contextual prompting	Provides the technical content and target audience.	Simplifies machine learning for beginners.
Few-shot prompting	Uses examples to demonstrate the expected style.	Shows how technical terms can be explained through everyday examples.
Constraint-based prompting	Defines rules and limitations.	Preserves accuracy and prevents unsupported information.
Structured-output prompting	Specifies the organization of the response.	Produces headings, definitions, examples, and summaries.
Audience-based prompting	Adapts content to the learner's knowledge level.	Avoids assuming advanced technical knowledge.

4. GENERATED OUTPUT

Topic: Machine Learning

A. Simple Introduction

Machine learning is a technology that helps computers learn from data and make predictions. It is a part of Artificial Intelligence (AI).

B. Explanation in Simple Language

Normally, a computer follows instructions given by a programmer. In machine learning, the computer is trained using examples and data so it can identify patterns and use them to make predictions.

For example, instead of manually writing rules for every spam email, we can train a machine-learning model using examples of spam and normal emails.

C. Important Technical Terms

Technical term	Simple meaning
Artificial Intelligence (AI)	Technology that enables computers to perform tasks requiring human-like intelligence.
Algorithm	A set of steps used to solve a problem.
Data	Information used by a computer.
Training data	Examples used to teach a machine-learning model.
Model	A system trained to recognize patterns and make predictions.
Prediction	An estimate of an outcome based on available information.

D. Step-by-Step Working Process

Step 1: Collect Data

Gather examples, such as spam and normal emails.

Step 2: Train the Model

The algorithm learns patterns from the examples.

Step 3: Identify Patterns

The model learns which characteristics are associated with spam.

Step 4: Make Predictions

The model predicts whether a new email is spam.

E. Real-World Example

Consider a student learning to identify different fruits.

- The student sees many apples and oranges.
- The student observes their color, shape, and size.
- After seeing enough examples, the student can identify a new fruit.

Similarly, a machine-learning model learns patterns from training data and uses them to classify new examples. Unlike a human, it learns through mathematical methods rather than conscious understanding.

F. Practical Applications

1. Email spam detection.
2. Movie and music recommendations.
3. Fraud detection in banking.
4. Voice recognition in smartphones.
5. Product recommendations in online shopping.
6. Identifying objects in images.

G. Advantages

- Helps automate repetitive tasks.
- Finds patterns in large amounts of data.
- Can improve performance through training.
- Supports faster predictions and decision-making.

H. Limitations

- Requires suitable, sufficiently representative data.
- Can produce incorrect predictions.
- May learn biases present in its training data.
- Requires evaluation and monitoring.

I. Common Beginner Mistakes

- Assuming AI and machine learning mean exactly the same thing.
- Thinking machine-learning models always produce correct results.
- Believing that a model can learn effectively from any data.
- Confusing training data with the predictions produced by a model.

J. Short Summary

Machine learning is a branch of AI that enables computers to learn patterns from data and use those patterns to make predictions. It is used in spam detection, recommendations, banking, and many other applications.

K. Review Questions

1. What is machine learning?

Answer: A branch of AI that learns patterns from data.

2. What is training data?

Answer: Examples used to train a machine-learning model.

3. Give one real-world application of machine learning.

Answer: Email spam detection.

5. WORKING PROCESS OF THE PROMPT

Input: Complex Technical Content

A paragraph, definition, formula, or technical explanation.

Prompt Engineering

Define the audience, role, simplification rules, and output structure.

Generative AI and LLM Processing

Interpret the input and generate a simpler explanation using the provided context.

Accuracy and Clarity Review

Check whether the important meaning is preserved and the language is suitable for beginners.

Output: Beginner-Friendly Explanation

Simple definitions, steps, examples, applications, and a summary.

6. AI CONCEPTS INVOLVED

Concept	Role in the task	Example
Artificial Intelligence (AI)	Broad field that enables machines to perform intelligent tasks.	AI-powered educational assistants.
Machine Learning (ML)	Learns patterns from data.	Personalizing learning recommendations.
Deep Learning (DL)	Uses multilayer neural networks to learn complex patterns.	Processing and understanding language.
Generative AI	Generates new content based on a prompt.	Rewriting a technical paragraph in simpler language.
Large Language Models (LLMs)	Process and generate human language based on context.	Explaining a programming concept to a beginner.

Relationship: AI is the broad field; ML is a subset of AI; DL is a subset of ML. Generative AI is an approach for generating content, often using deep-learning models. LLMs are deep-learning models specialized in language tasks and can be used for text simplification.

7. EVALUATION BASED ON THE RUBRICS

Criterion	**Weightage**	**How the prompt meets it**
Technique Selection	30%	Uses role-based, instruction-based, contextual, audience-based, and structured-output prompting.
Output Effectiveness	25%	Includes clear definitions, real-world examples, practical applications, and review questions.
Prompt Craftsmanship	20%	Organizes the instructions into a logical sequence with explicit requirements.
Prompt-Output Documentation	15%	Presents the prompt, generated output, workflow, and examples using headings and tables.
Design Rationale	10%	Explains why each instruction improves clarity, accuracy, and beginner understanding.
Total	100%	Addresses all five evaluation criteria.

8. DESIGN RATIONALE

The prompt is designed to make technical learning easier without losing essential meaning.

- Clear role: Encourages the AI to explain concepts as a teacher would.
- Target audience: Prevents the use of unexplained advanced terminology.
- Step-by-step explanations: Breaks complicated information into manageable parts.
- Real-world examples: Connects abstract ideas with familiar experiences.
- Output structure: Makes the response easy to read, revise, and remember.
- Accuracy constraints: Reduce the risk of changing definitions or introducing false information.
- Review questions: Help learners check their understanding.

9. RESULT

A detailed prompt was designed to convert complex technical content into simple language for beginners using Generative AI and LLMs. The prompt combines multiple prompt-engineering techniques, real-world examples, structured explanations, and accuracy constraints. The resulting content is easier to understand, remember, and apply, making the approach useful for students, teachers, technical documentation, and self-learning.